Executive_Summary_Report

Project Title: ERP Sales Data Analysis and Business Intelligence Dashboard

Introduction:

The objective of this project was to clean, analyze, and visualize an ERP sales dataset using Microsoft Excel, SQL Server, and Microsoft Power BI. The project focused on improving data quality, organizing business data, performing analytical queries, and creating an interactive dashboard to support business decision-making.

Project Objectives:

The main objectives of this project were:

- To identify and resolve data quality issues.
- To clean and prepare the dataset using Microsoft Excel Power Query.
- To design a structured SQL database for storing ERP sales data.
- To perform SQL analytical queries for business insights.
- To build an interactive Power BI dashboard for visual reporting.
- To support data-driven decision-making through business intelligence.

Tools and Technologies Used:

- Microsoft Excel
- Power Query
- SQL Server
- SQL Server Management Studio (SSMS)
- Microsoft Power BI

Work Performed:

During this project, the following tasks were completed:

- Performed data profiling to assess data quality.
- Removed duplicate records.
- Cleaned and standardized text values.
- Removed invalid email addresses.
- Corrected inconsistent country names.
- Filtered invalid Quantity and UnitPrice values.

- Created calculated columns including TotalSales, OrderYear, and OrderMonth.
- Developed SQL database schema.
- Executed SQL analytical queries to generate business insights.
- Designed an interactive Power BI dashboard containing sales trends, regional analysis, product performance, and customer insights.

Key Outcomes:

The project successfully transformed raw ERP sales data into a clean and structured dataset suitable for analysis. SQL queries provided valuable business insights, while the Power BI dashboard enabled interactive visualization of sales performance, customer behavior, and regional trends. The final solution improved data accuracy, reporting efficiency, and decision-making capabilities.

Conclusion:

This project demonstrated the complete workflow of a data analytics solution, including data cleaning, database development, SQL analysis, and dashboard visualization. The final deliverables provide a reliable foundation for business reporting and future analytical activities.